CSA0921 - Programming in Java
CO3 - AT1 - Debugging and Optimization

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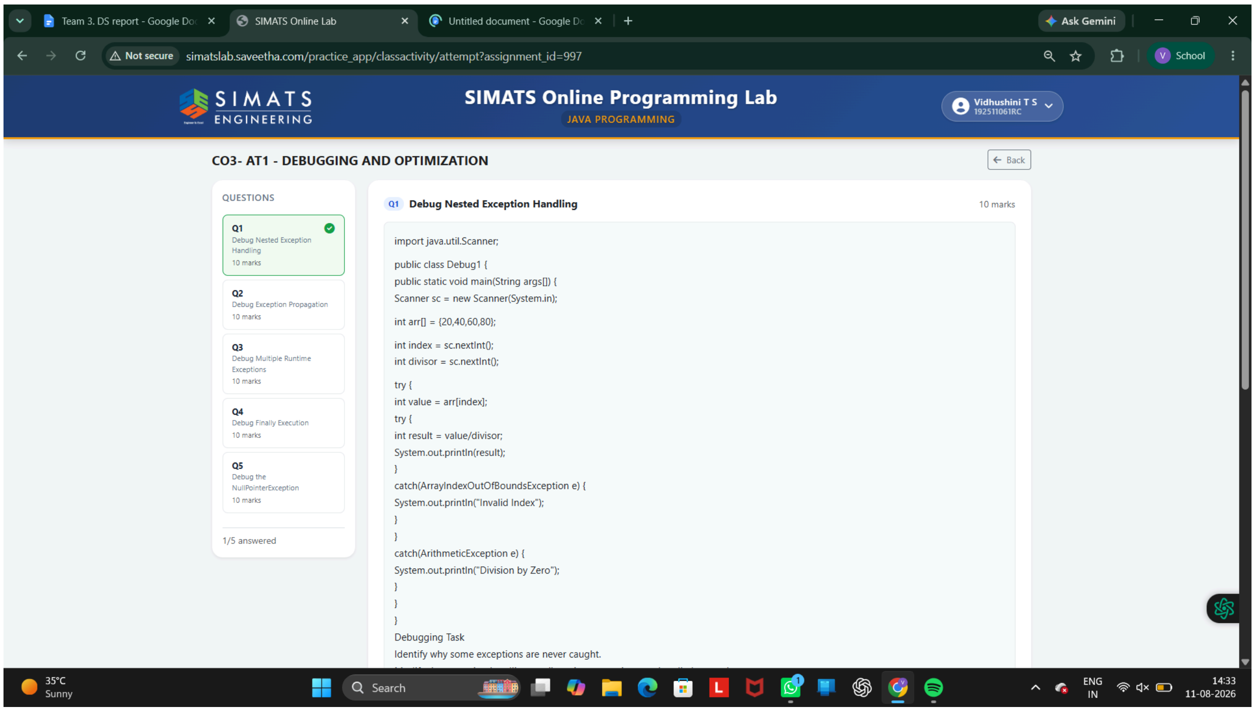
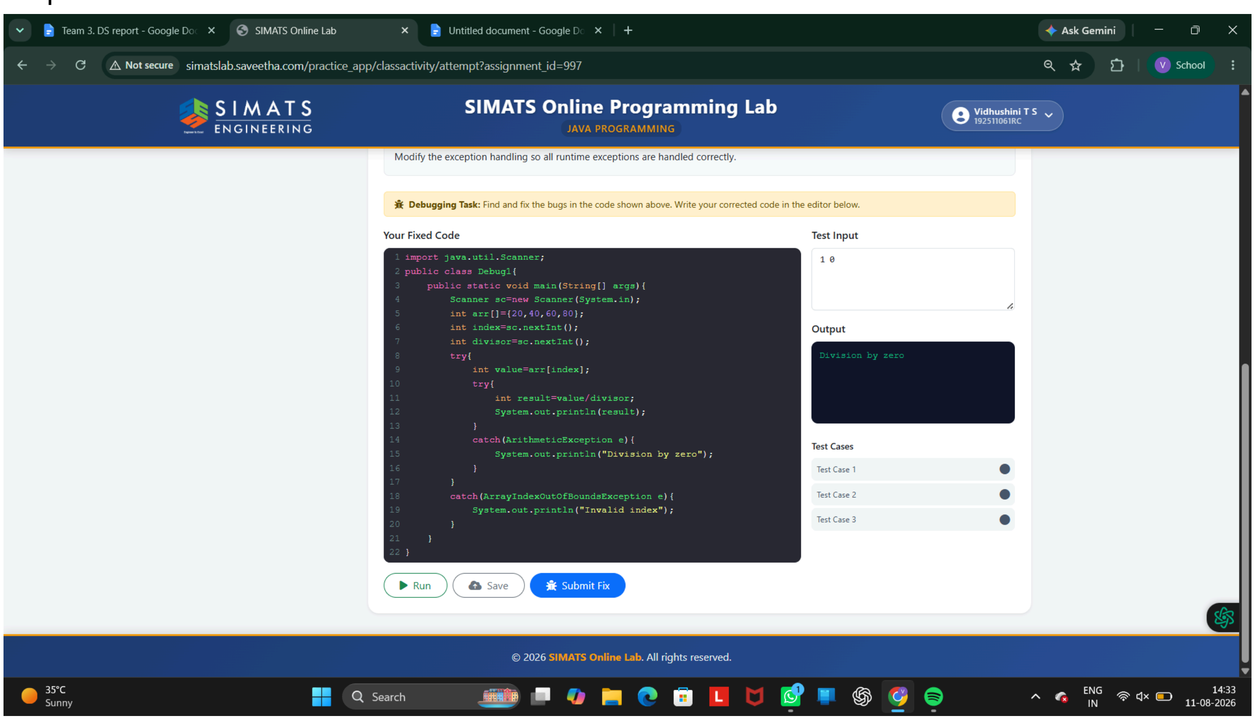
Reg. No.: 192511061

1.



Output:



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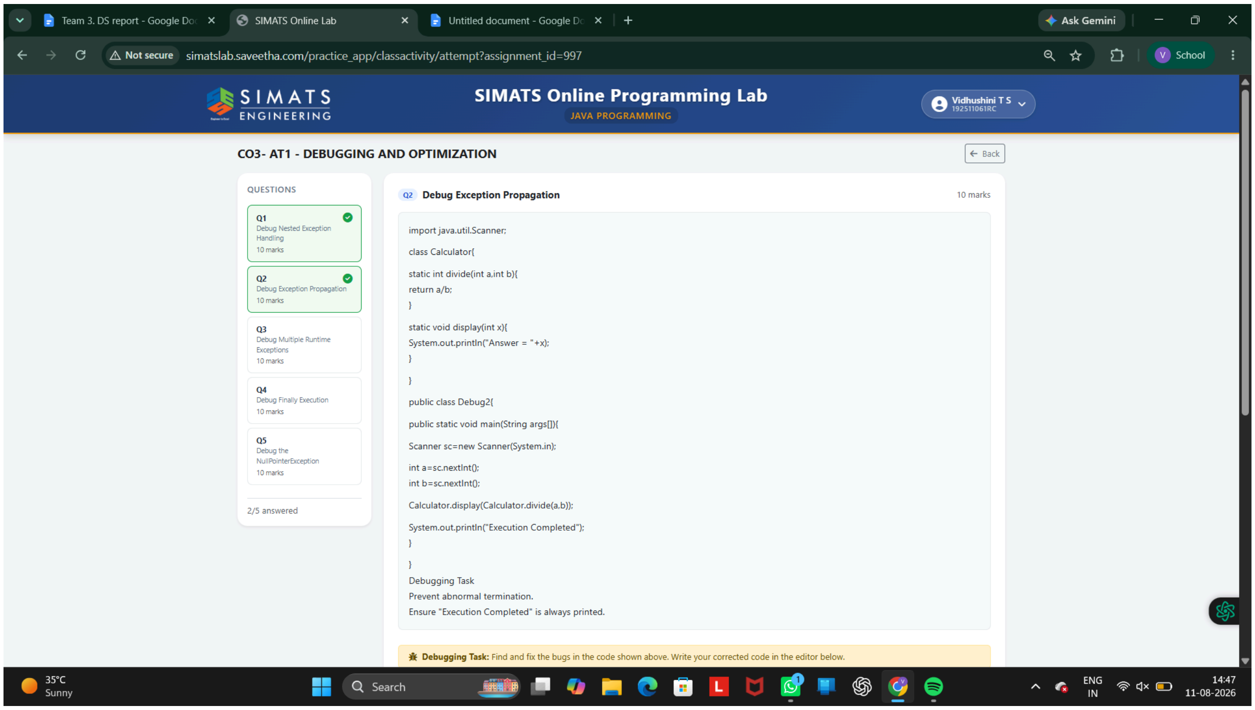
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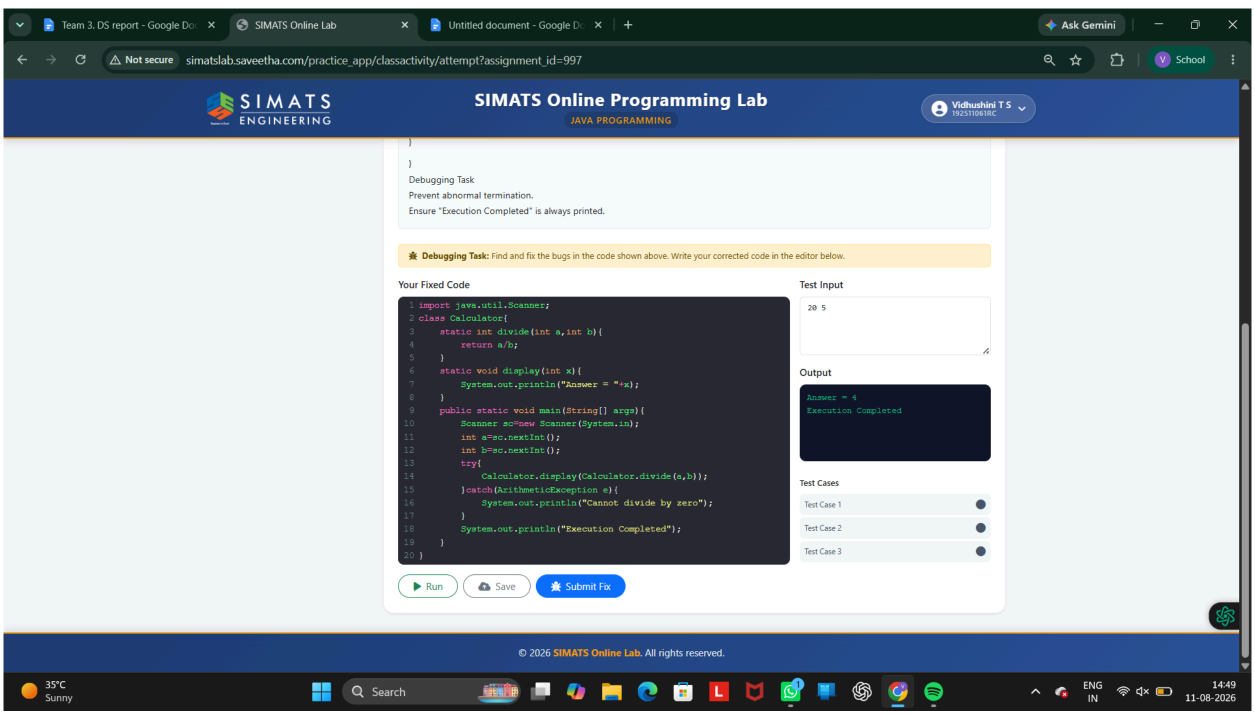
CO3- AT1 - DEBUGGING AND OPTIMIZATION
QUESTIONS
Q1 Debug Nested Exception Handling 10 marks
Q2 Debug Exception Propagation 10 marks
Q3 Debug Multiple Runtime Exceptions 10 marks
Q4 Debug Finally Execution 10 marks
Q5 Debug the NullPointerException 10 marks
2/5 answered
Q2 Debug Exception Propagation
10 marks
import java.util.Scanner;
class Calculator{
static int divide(int a,int b){
return a/b;
}
static void display(int x){
System.out.println("Answer = "+x);
}
}
public class Debug2{
public static void main(String args[]){
Scanner sc=new Scanner(System.in);
int a=sc.nextInt();
int b=sc.nextInt();
Calculator.display(Calculator.divide(a,b));
System.out.println("Execution Completed");
}
}
Debugging Task
Prevent abnormal termination.
Ensure "Execution Completed" is always printed.
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Output: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Debugging Task
Prevent abnormal termination.
Ensure "Execution Completed" is always printed.
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
import java.util.Scanner;
class Calculator{
static int divide(int a,int b){
return a/b;
}
static void display(int x){
System.out.println("Answer = "+x);
}
public static void main(String[] args){
Scanner sc=new Scanner(System.in);
int a=sc.nextInt();
int b=sc.nextInt();
try{
Calculator.display(Calculator.divide(a,b));
}catch(ArithmeticException e){
System.out.println("Cannot divide by zero");
}
System.out.println("Execution Completed");
}
}
Test Input
20 5
Output
Answer = 4
Execution Completed
Test Cases
Test Case 1
Test Case 2
Test Case 3
Run
Save
Submit Fix
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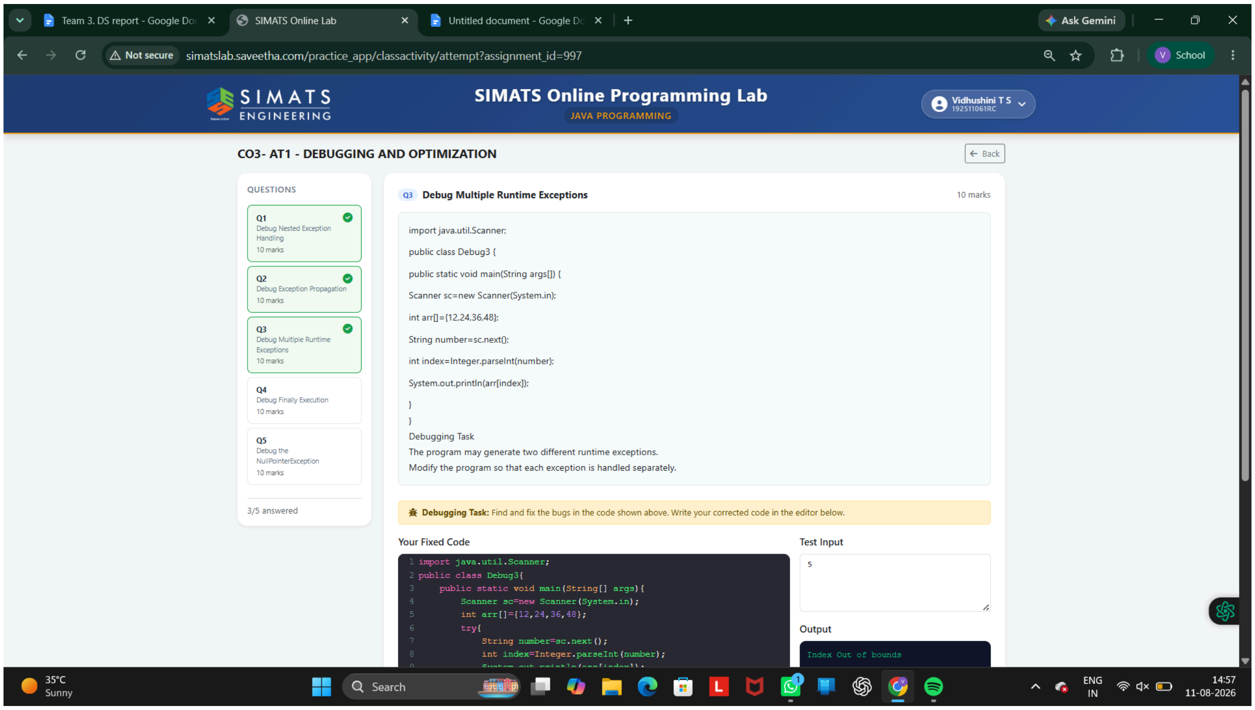
3.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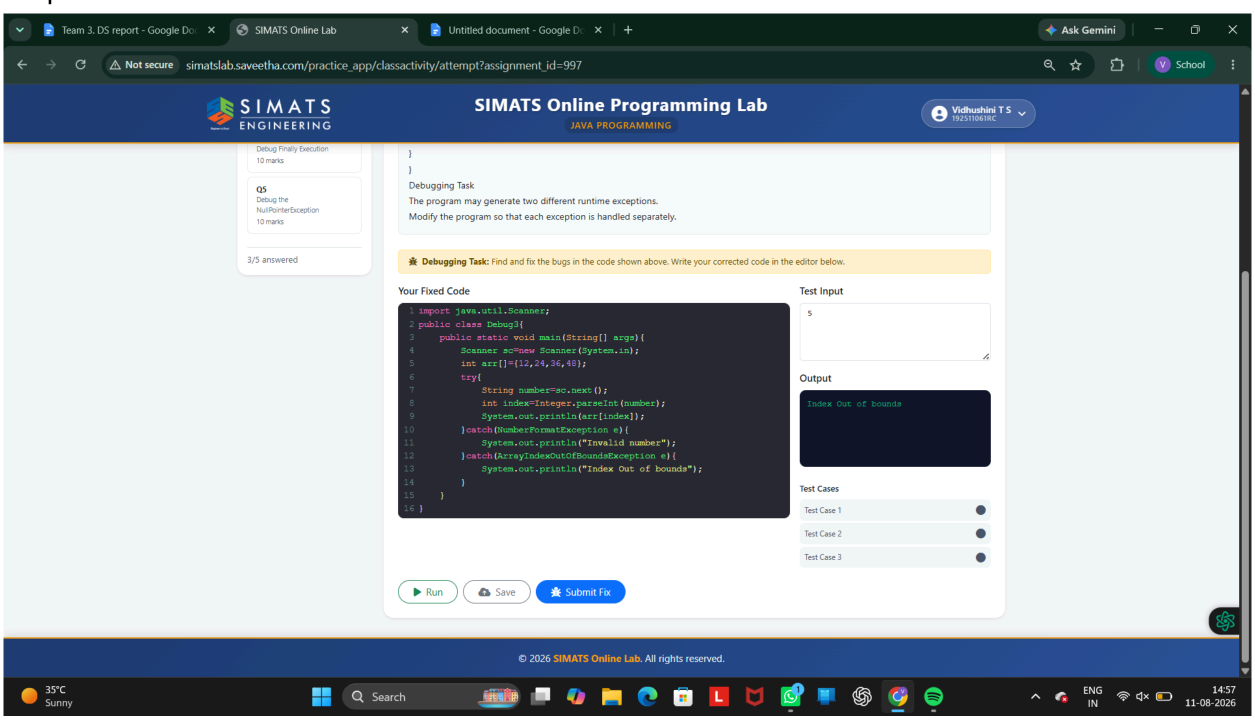
CO3- AT1 - DEBUGGING AND OPTIMIZATION
Q3 Debug Multiple Runtime Exceptions
10 marks
import java.util.Scanner;
public class Debug3 {
public static void main(String args[]) {
Scanner sc=new Scanner(System.in);
int arr[]={12,24,36,48};
String number=sc.next();
int index=Integer.parseInt(number);
System.out.println(arr[index]);
}
}
Debugging Task
The program may generate two different runtime exceptions.
Modify the program so that each exception is handled separately.
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
Test Input
Output
Index Out of bounds
3/5 answered

Output: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Debugging Task
The program may generate two different runtime exceptions.
Modify the program so that each exception is handled separately.
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
import java.util.Scanner;
public class Debug3{
public static void main(String[] args){
Scanner sc=new Scanner(System.in);
int arr[]={12,24,36,48};
try{
String number=sc.next();
int index=Integer.parseInt(number);
System.out.println(arr[index]);
}catch(NumberFormatException e){
System.out.println("Invalid number");
}catch(ArrayIndexOutOfBoundsException e){
System.out.println("Index Out of bounds");
}
}
}
Test Input
5
Output
Index Out of bounds
Test Cases
Test Case 1
Test Case 2
Test Case 3
Run
Save
Submit Fix
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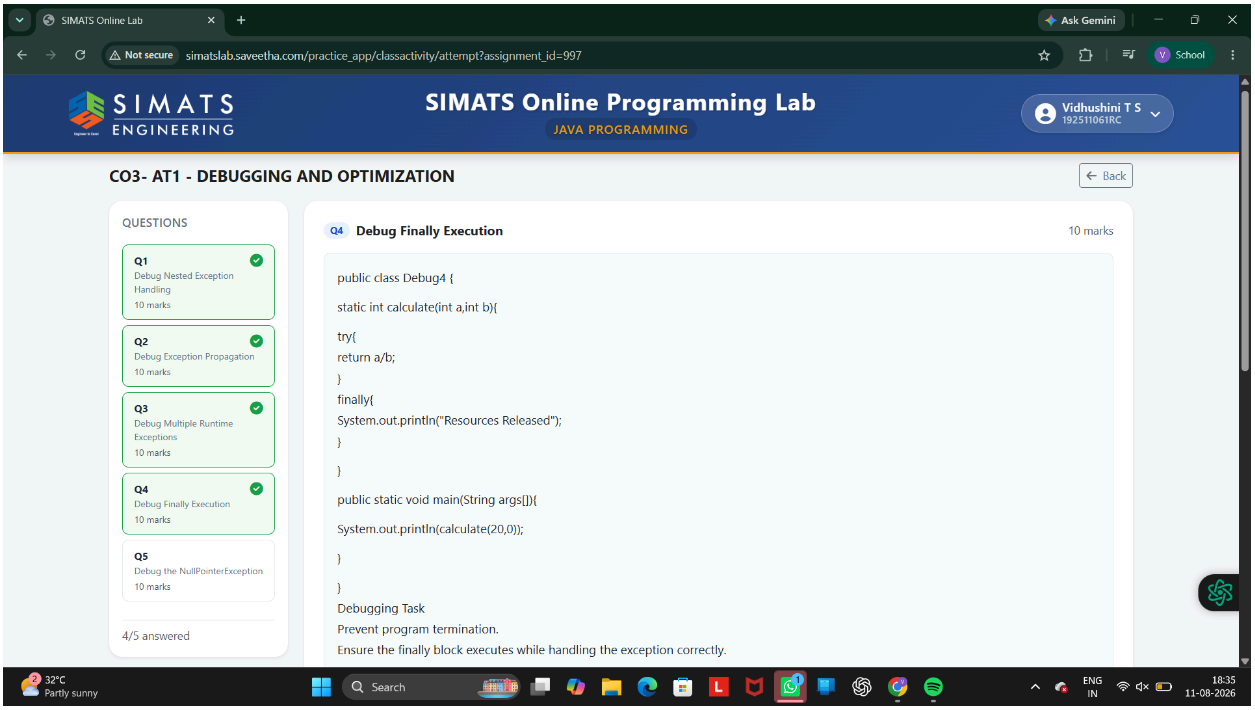
4.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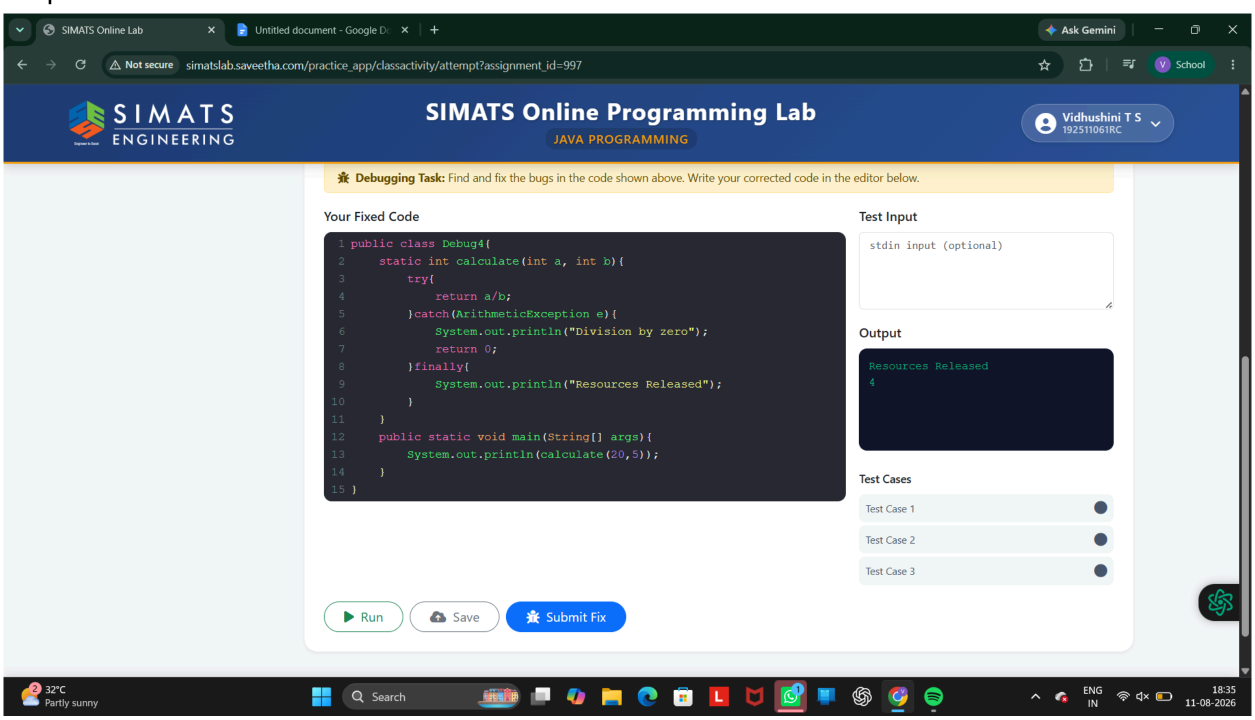
Back
QUESTIONS
Q1
Debug Nested Exception Handling
10 marks
Q2
Debug Exception Propagation
10 marks
Q3
Debug Multiple Runtime Exceptions
10 marks
Q4
Debug Finally Execution
10 marks
Q5
Debug the NullPointerException
10 marks
4/5 answered
Q4
Debug Finally Execution
10 marks
public class Debug4 {
static int calculate(int a,int b){
try{
return a/b;
}
finally{
System.out.println("Resources Released");
}
}
public static void main(String args[]){
System.out.println(calculate(20,0));
}
}
Debugging Task
Prevent program termination.
Ensure the finally block executes while handling the exception correctly.
32°C
Partly sunny
Search
ENG
IN
18:35
11-08-2026

Output: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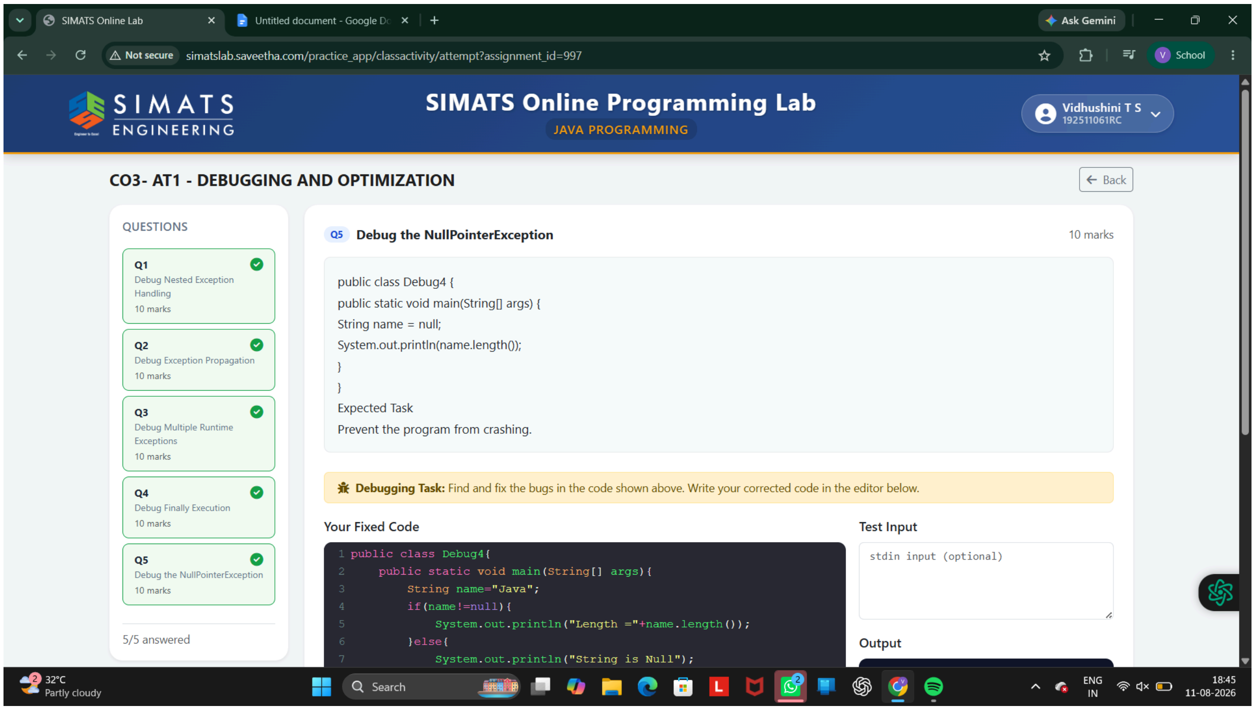
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
public class Debug4{
static int calculate(int a, int b){
try{
return a/b;
}catch(ArithmeticException e){
System.out.println("Division by zero");
return 0;
}finally{
System.out.println("Resources Released");
}
}
public static void main(String[] args){
System.out.println(calculate(20,5));
}
}
Test Input
stdin input (optional)
Output
Resources Released
4
Test Cases
Test Case 1
Test Case 2
Test Case 3
Run
Save
Submit Fix
32°C
Partly sunny
Search
ENG
IN
18:35
11-08-2026

5.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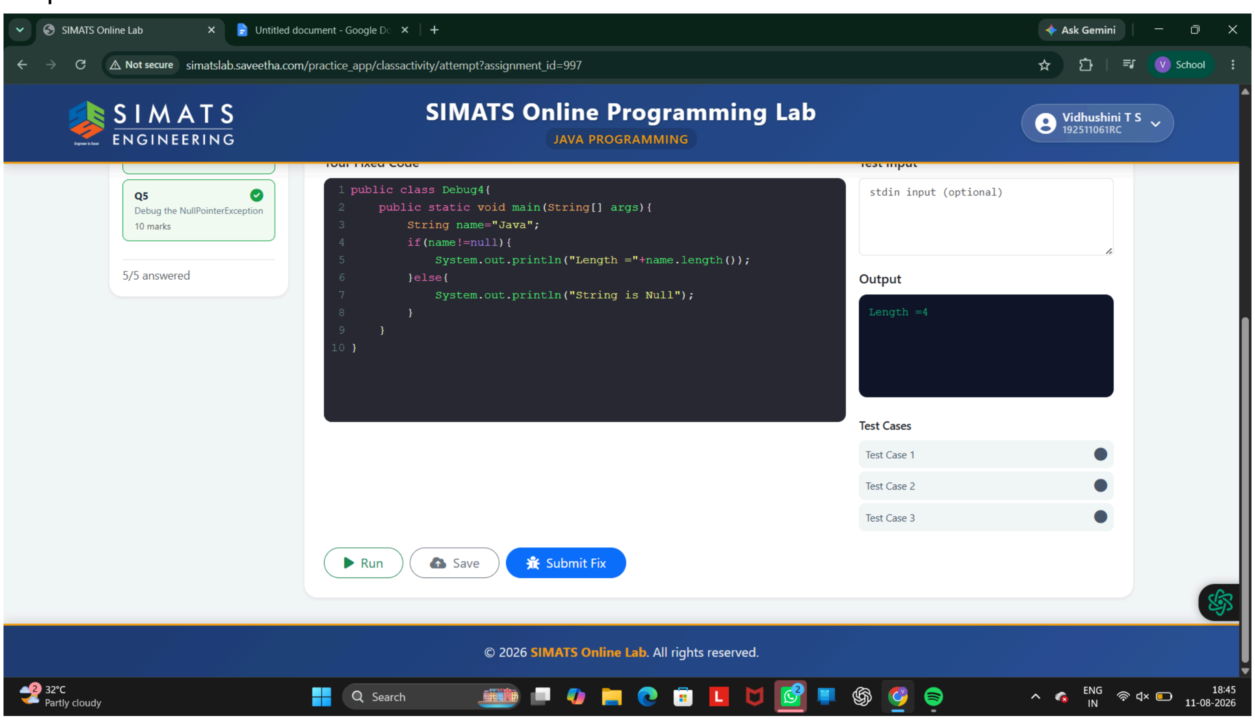
QUESTIONS
Q1 Debug Nested Exception Handling 10 marks
Q2 Debug Exception Propagation 10 marks
Q3 Debug Multiple Runtime Exceptions 10 marks
Q4 Debug Finally Execution 10 marks
Q5 Debug the NullPointerException 10 marks
5/5 answered
Q5 Debug the NullPointerException
10 marks
public class Debug4 {
public static void main(String[] args) {
String name = null;
System.out.println(name.length());
}
}
Expected Task
Prevent the program from crashing.
Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.
Your Fixed Code
public class Debug4{
public static void main(String[] args){
String name="Java";
if(name!=null){
System.out.println("Length ="+name.length());
}else{
System.out.println("String is Null");
Test Input
stdin input (optional)
Output
32°C
Partly cloudy
Search
ENG
IN
18:45
11-08-2026

Output: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Q5 Debug the NullPointerException 10 marks
5/5 answered
public class Debug4{
public static void main(String[] args){
String name="Java";
if(name!=null){
System.out.println("Length ="+name.length());
}else{
System.out.println("String is Null");
}
}
}
stdin input (optional)
Output
Length =4
Test Cases
Test Case 1
Test Case 2
Test Case 3
Run
Save
Submit Fix
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32°C
Partly cloudy
Search
ENG
IN
18:45
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